

HOW TO ENCODE AND DECODE INFORMATION

Learn to convert between various codes for digital data transmission



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Digital data is represented, stored and exchanged in the form of codes written in strings of 0's and 1's known as bits. And encoding and decoding is an art of transforming one representation into another and vice versa. In accounts section of offices, employee codes are used for salary disbursement. The code is a unique number given to the employee name. Decoding is getting the employee details from the code.

In computer engineering, encoding and decoding are used for transformation of numerical data from one system to another system for specific

use. There are different kinds of binary codes, like weighted and non-weighted codes, reflected codes, sequential codes, alphanumeric codes, and error detection and correction codes.

Weighted code

Converting decimal data into binary and vice versa is essential for computing as the computer realises only binary data from 'on' and 'off' states of the transistor built into the system. Binary codes are the codes that are represented in binary system with modification from the original ones. There are two such codes: weighted binary codes and non-weighted codes.

In addition to conversion of the number system, it is often desirable to retain decimal character even when it is encoded in binary digits, or bits (1 or 0). For example, consider decimal number 56, where we represent individual decimal digits, namely, 5 and 6, as 0101 for 5 and 0110 as 6:

56=01010110

Such a representation is known as binary-coded decimal (BCD). It is a code known as weighted code (see Table I).

From Table I, you can find the code of 783 as 0111 1000 0011. In fact, each digit of the decimal number is coded as 4-tuple binary data, say, $a_3a_2a_1a_0$, as follows:

First decimal digit	Second decimal digit	Last decimal digit
$a_3a_2a_1a_0$	$a_3a_2a_1a_0$	$a_3a_2a_1a_0$

BCD is also known as 8421 code, where 8, 4, 2 and 1, respectively, rep-

TABLE I
Weighted BCD Codes for Decimal Digits

Decimal digit	BCD code weight			
	8	4	2	1
0	0	0	0	00
1	0	0	0	01
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	01
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	0	0	0	01

resent weights of fourth, third, second and first positions of the tuple.

As examples of other weighted codes, Table II shows following codes: 2421, 74-2-1 and 4221.

In fact, weighted binary codes are those which obey the positional weighting principle (each position of the number represents a specific weight).

Some weighted codes are known as reflective codes. A code is said to be reflective when the code for 9 is the complement for the code for 0, code for 8 is the complement for the code for 1, and so on. Codes 2421, 5211 and excess-3 are reflective, whereas the 8421 code is not reflective.

Similarly, some weighted codes are sequential. A code is said to be sequential when two subsequent codes, seen as numbers in binary representation, differ by one. This greatly aids in mathematical manipulation of data. The 8421 and Excess-3 codes are